

Acoustically driven coupled-ring single-photon frequency shifter on thin-film lithium niobate: a design study

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We present a simulation-only design study of a single-photon frequency shifter on thin-film lithium niobate (TFLN) in which a suspended interdigital-transducer (IDT) acoustic resonator, instead of electrodes, drives one ring of a coupled-ring photonic molecule. We integrate the full time-domain coupled-mode equations without the rotating-wave approximation, with the drive calibrated to a measured resonant acousto-optic modulator, and report the design under realistic, partly assumed conditions: 10% IDT efficiency, an acoustic quality factor $Q_m = 2000$ (measured maximum 3600 as upper bound; an unsourced, assumed 1570 as conservative case), RF retuning limited to ± 1 MHz by the acoustic linewidth, extra optical loss from the IDT and suspended region, residual temperature jitter after locking, and coupling-gap fabrication errors. In a 1000-sample Monte Carlo the median fraction of output light shifted into the single sideband 3.33 GHz away is $\eta_{\text{shift}} = 94.3\%$ (10th–90th percentile 82.4–97.9%), with 44.7% of samples at $\geq 95\%$; the median on-chip insertion loss is 5.0 dB, the on-chip absolute efficiency 28.4% (21.5–38.8%), and the fiber-to-fiber efficiency 14.2% for an assumed 1.5 dB per facet, at 33.7 μW of source RF power for $Q_m = 2000$ (18.7 μW at $Q_m = 3600$, 43 μW at the assumed $Q_m = 1570$). The idealized operating point ($\eta_{\text{shift}} = 98.3\%$, 55% absolute efficiency, 6.7 μW) is an upper bound. For comparison, the best measured acousto-optic shifter on TFLN, a broadband traveling-wave device, reaches 3.5% on-chip efficiency at 1 W. The design does *not* offer a power advantage over impedance-matched electro-optic (EO) driving, its usable RF bandwidth is only ~ 1 MHz, and it needs two demanding controls: active resonance locking at the ± 9.8 mK level (thermo-optic effect plus thermal expansion), and coupling-gap control within ≈ 5 nm to hold the supermode splitting at the acoustic frequency. For a quantum interface, the Hong–Ou–Mandel (HOM) visibility with a delay-matched reference photon and a target-bin filter is $1 - V = 8 \times 10^{-7}$ for a 10-MHz-bandwidth photon and 6×10^{-4} at 50 MHz at the ideal point (unfiltered, $V \approx \eta_{\text{shift}}$), and thermal phonons contribute a negligible incoherent coupling noise ($n_{\text{th}}/N_{\text{coh}} \approx 10^{-8}$) at room temperature.

I. INTRODUCTION

Photonic qubits encoded in discrete frequency bins need low-loss, coherent devices that move single photons between bins [1–3]. TFLN combines strong Pockels, piezoelectric, and photoelastic responses with low-loss resonators [4, 5], and has been used for single-photon spectral shearing over ± 641 GHz [6], an integrated quantum frequency processor [7], and coupled-cavity frequency beam splitters [8]. For single-tone shifting, the benchmark is the EO coupled-ring shifter of Hu *et al.* [9]: a microwave drive couples the symmetric (S) and antisymmetric (AS) supermodes of a photonic molecule [10] and, at a generalized critical-coupling condition, transfers light unidirectionally from one supermode to the other. The measured device shifts 98.7% of the output light by 28.2 GHz with 0.45 dB insertion loss (IL), and 99.1% (up) and 99.4% (down) at 12.5 GHz, using 96 mW of microwave power [9]. A different EO approach, serrodyne driving of cascaded traveling-wave TFLN phase modulators, reports a shifting efficiency above 98.8% with $V_\pi \approx 2.3$ V [11].

Acoustic frequency shifting on TFLN is much less mature. The best published TFLN acousto-optic frequency shifter, a traveling-wave Bragg-deflection device at ~ 3 GHz, reaches an on-chip efficiency of 3.5% at 30 dBm (1 W) applied to the IDT pads, with > 30 dB

carrier suppression and a 70-MHz microwave bandwidth [12]. Resonant acoustics improves the per-volt modulation strongly: a suspended 100- μm acoustic resonator at 3.33 GHz gives $V_\pi = 4.6$ V ($V_\pi L = 0.046$ V cm) on one arm of a Mach–Zehnder interferometer (MZI) [13]. Acoustic waves have long been used to modulate integrated resonators [14] and to control photonic molecules dynamically [15].

Most relevant here, Zhu *et al.* [16] have already driven a lithium niobate (LN-on-sapphire) coupled-microring photonic molecule acoustically. In their device an IDT-launched 8.54-GHz acoustic wave forms a backward-Brillouin dynamic Bragg mirror that couples forward- and backward-propagating light, reaching strong supermode coupling at milliwatt-level acoustic power (cooperativity 2.46 mW^{-1}) and Bragg reflectivities up to 24%. Acoustic driving of an LN photonic molecule is therefore not new. The distinction of the present design is its operating point: a *forward* (co-directional), *single-bus* frequency shifter that sends $\approx 98\%$ of the output into one sideband, analyzed in terms of shift efficiency, insertion loss, spurious orders, and the RF power at the source. We also considered an all-optical, IDT-free alternative in which backward stimulated Brillouin scattering in TFLN, pumped by an EO frequency comb, writes the phonon; in our (unpublished) analysis it is not viable for multi-bin frequency operations, whereas forward-Brillouin frequency-bin swaps have been proposed theo-

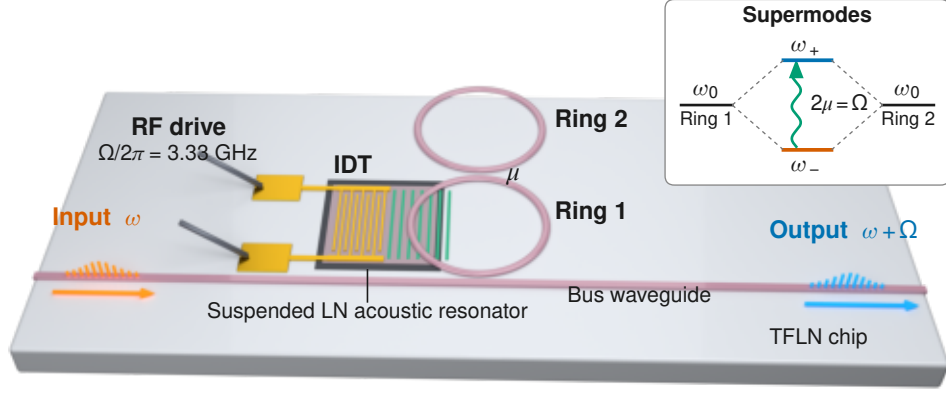


FIG. 1. Device concept. An RF-driven interdigital transducer (IDT; $\Omega/2\pi = 3.33$ GHz) excites a suspended LN acoustic resonator that covers only a segment of ring 1. Ring 2 is coupled only to ring 1 (coupling rate μ), and a single bus waveguide is coupled only to ring 1. Light enters in the forward direction at ω and leaves the same bus at the shifted frequency $\omega + \Omega$. Inset: the ring–ring coupling splits the two degenerate ring modes (ω_0) into supermodes $\omega_{\pm}$ separated by $2\mu = \Omega$, which the acoustic drive couples. Schematic, not to scale.

retically [17],[18]

In this paper we (i) define the device (Fig. 1) and calibrate its drive to measured data [Sec. II]; (ii) compute its operating point and compare its source power with an EO drive of identical optics, *including* impedance matching on both sides [Sec. III]; (iii) quantify its tolerance to optical Q , RF detuning, IDT efficiency, thermal drift, and ring mismatch, and evaluate it under realistic parameters with a Monte Carlo analysis [Secs. IV and V]; and (iv) evaluate its quantum performance through the HOM visibility and thermal-phonon noise [Sec. VI]. Limitations are collected in Sec. VII. All device numbers in this paper are simulation results; none are measurements.

II. DEVICE AND MODEL

Structure.—The device (Fig. 1) is the coupled-ring photonic molecule of Ref. [9], with only ring 1 coupled to the bus waveguide. A $100\text{-}\mu\text{m}$ section of ring 1 passes through a suspended LN acoustic resonator driven by an IDT, with the geometry and calibration of Ref. [13]. The ring length is $L_{\text{ring}} = 0.667$ mm, so the acoustic field covers 15% of the ring. This is the same coverage ratio, $\eta_{\text{cav}} = 0.15$, as the suspended racetrack of Ref. [13]. We take $n_g = 2.3$. The ring–ring coupling is chosen so that the doublet splitting equals the acoustic frequency, $2\mu = \Omega_m = 2\pi \times 3.33$ GHz.

Equations of motion.—We use Eq. (S2) of Ref. [9], with ring 1 coupled to the bus at rate γ and both rings having intrinsic loss rate κ_i :

$$\begin{aligned} \dot{a}_1 &= \left[-i(\Delta + \delta\omega \cos \Omega_m t) - \frac{\gamma + \kappa_i}{2} \right] a_1 - i\mu a_2 - \sqrt{\gamma} \alpha_{\text{in}}, \\ \dot{a}_2 &= \left[-i(\Delta + s\delta\omega \cos \Omega_m t) - \frac{\kappa_i}{2} \right] a_2 - i\mu a_1, \\ a_{\text{out}} &= \alpha_{\text{in}} + \sqrt{\gamma} a_1. \end{aligned} \quad (1)$$

For the acoustic (AO) drive only ring 1 is modulated ($s = 0$); for the EO reference we use Hu’s push-pull drive ($s = -1$). The laser is tuned to the S supermode ($\Delta = \mu$). We integrate Eq. (1) with a fourth-order Runge–Kutta scheme (80 steps per modulation period, 40 ns total) without the rotating-wave approximation, and project a_{out} , averaged over the final 20 ns, onto the frequencies $\omega_L + q\Omega_m$ ($|q| \leq 3$). In the S/AS basis the single-ring term contains $\frac{\delta\omega}{2} \cos \Omega_m t (c_S^\dagger c_A + \text{h.c.})$ plus a common-mode phase modulation of both supermodes, so the supermode coupling is $\delta\omega/2$. As in Ref. [9], we define $\eta_{\text{shift}} = P_{\text{shift}}/P_{\text{out}}$, the absolute efficiency $\eta_{\text{abs}} = P_{\text{shift}}/P_{\text{in}}$, and $\text{IL} = -10 \log_{10}(P_{\text{out}}/P_{\text{in}})$.

Drive calibration.—A peak voltage V from a $50\text{-}\Omega$ source ($P_{\text{RF}} = V^2/2R$, $R = 50 \text{ }\Omega$) produces the phase $\Delta\phi = \pi V L_a / (V_\pi L)$ over the covered length $L_a = 100 \text{ }\mu\text{m}$, which shifts the ring resonance by

$$\delta\omega = \Delta\phi \frac{c}{n_g L_{\text{ring}}}. \quad (2)$$

With $V_\pi L = 0.046 \text{ V cm}$ [13] this gives $\delta\omega/2\pi = 21.3 \text{ GHz/V}$. Reference [13] extracts V_π from the measured S_{21} referenced to the power of a $50\text{-}\Omega$ source, so this calibration already contains the measured $\approx 50\%$ electrical-to-acoustic coupling of that IDT (a -3 dB dip in S_{11}). We refer to this case as the “Shao IDT”; an ideally matched IDT would need half the power, and an IDT of efficiency η_{IDT} needs $0.5/\eta_{\text{IDT}}$ times the Shao-IDT power. For the acoustic resonance we use $Q_m = 2000$ and $\gamma_e/\gamma = 0.15$, the values used to model the measured devices in Ref. [13] (measured acoustic Q up to 3600, Fig. 3(c) of Ref. [13]).

Optical parameters.—We take $\gamma/2\pi = 0.6 \text{ GHz}$ and the intrinsic loss implied by the suspended racetrack of Ref. [13]: from its Table S2, $\kappa/2\pi = 95 \text{ MHz}$ and $2\kappa_e/\kappa = 0.3$, so $\kappa_i/2\pi = 0.85 \times 95 \text{ MHz} = 80.75 \text{ MHz}$, i.e. $Q_i \approx 2.4 \times 10^6$ (at 193.4 THz). The main text of Ref. [13]

quotes a loaded Q of 2.2×10^6 for the same linewidth.

EO reference with identical optics.—The EO drive uses the same optical parameters, push-pull modulation, and Hu’s EO coefficient of ≈ 0.5 GHz/V referred to the voltage V_c on the electrode capacitor [9]. We model the electrode as Hu’s lumped circuit, $C = 0.11$ pF, $L = 0.12$ nH, $R = 1.5$ Ω [9], whose reactance at 3.33 GHz is $X_C = 434$ Ω . We consider four feeds: (a) $V_c = V_0$, the convention of Ref. [9], with V_0 the source peak voltage; (b) an open-circuited capacitor, $V_c = 2V_0$; (c) a lumped LC match with inductor quality factor $Q_L = 30$ or 100, for which $P_{\text{RF}} = V_c^2(R + X_C/Q_L)/(2X_C^2)$; and (d) an ideal inductor, $P_{\text{RF}} = V_c^2 R/(2X_C^2)$. The $Q_L = 30$ match has an RF bandwidth of ≈ 122 MHz.

III. OPERATING POINT AND DRIVE POWER

Ideal operating point (upper bound).—Sections III and IV use nominal, idealized parameters (identical rings, no fabrication error, no extra loss, perfect temperature control); the realistic-condition results that we regard as the main prediction are given in Sec. V. At $\gamma/2\pi = 0.6$ GHz the acoustic drive reaches its maximum $\eta_{\text{shift}} = 98.3\%$ at $\delta\omega/2\pi = 0.55$ GHz, i.e. a source peak voltage of 26 mV and $P_{\text{RF}} = 6.7$ μW with the Shao IDT (Table I, Fig. 2). The absolute efficiency is 55% and the IL is 2.5 dB; both are set by the intrinsic loss rather than by the drive. The shifted line is the $q = +1$ order. Relative to it, the carrier is at -23.2 dB and the largest spurious order is $q = -1$ at -21.2 dB ($q = +2$ at -23.6 dB). These spurs come from the common-mode phase modulation of single-ring driving. The push-pull EO drive cancels this term and reaches $\eta_{\text{shift}} = 99.6\%$, with the carrier at -26.0 dB and the $q = -1$ order at -30.6 dB [Fig. 2(b)], and the same 2.5 dB IL.

Comparison with EO driving.—Table II lists the source power at the respective optima for each feed. Without any matching on the EO side ($V_c = V_0$) the EO drive needs 3.1 mW, ≈ 460 times the 6.7 μW of the acoustic drive, and the measured device of Ref. [9] used 96 mW. These factors, however, compare a resonantly enhanced acoustic drive with a non-resonant electrode, and they do not survive a like-for-like comparison. With an LC match the EO power falls to 13 μW ($Q_L = 30$), 4.8 μW ($Q_L = 100$), or 1.2 μW (ideal inductor). Against the ideally matched IDT (3.4 μW), the two drives therefore differ only by a factor of 0.36–3.9, and *the EO drive is the more power-efficient one when the inductor loss is low*. Against the as-measured Shao IDT (6.7 μW) the range is 0.18–1.9. We therefore make no claim of a power advantage over EO driving. The remaining differences are qualitative: the acoustic resonator provides voltage gain without an external matching network, but at the cost of a ~ 100 -fold narrower RF bandwidth (Sec. IV) and a fixed shift frequency.

Comparison with acousto-optic shifters.—The best measured TFLN acousto-optic shifter reaches 3.5% on-

chip efficiency at 30 dBm (1 W) [12]. The present design reaches, in simulation, $\eta_{\text{abs}} = 55\%$ ($\eta_{\text{shift}} = 98.3\%$) at 6.7 μW under ideal conditions and a median on-chip $\eta_{\text{abs}} = 28.4\%$ at 33.7 μW ($Q_m = 2000$, 10% IDT efficiency) under the realistic conditions of Sec. V. The comparison is not like for like: Ref. [12] is a broadband, non-resonant traveling-wave deflector with a 70-MHz microwave bandwidth, whereas our design is a simulated, doubly resonant (optical and acoustic) device with a ~ 1 -MHz RF bandwidth and a single fixed shift frequency; a measured device will also face the fabrication and locking requirements of Sec. V. Zhu *et al.* [16] operate an acoustic photonic molecule at milliwatt acoustic power in a backward, reflective geometry and do not report a forward shift efficiency or insertion loss. Table III summarizes these benchmarks.

IV. TOLERANCES

Optical quality factor.—Because the coupling γ is fixed while κ_i varies, the shift ratio is robust but the loss is not [Fig. 3(a)]. For $Q_i \gtrsim 1.3 \times 10^6$ we find $\eta_{\text{shift}} \approx 98\%$; at $Q_i = 6.4 \times 10^5$ it drops to 82%. The absolute efficiency rises from 33% at $Q_i = 1.3 \times 10^6$ through 55% at 2.4×10^6 to 87% at 9.7×10^6 (IL 0.56 dB). The required source power stays between 6.1 and 8.5 μW over this range.

RF detuning.—The acoustic resonance filters the drive. With $Q_m = 2000$ (FWHM 1.67 MHz) we find $\eta_{\text{shift}} \geq 96.7\%$ within ± 0.4 MHz of the doublet splitting, 80% at 1 MHz, 44% at 2 MHz, and 10% at 5 MHz [Fig. 3(b)]. The usable RF bandwidth is thus ~ 1 MHz. Under the flat-drive assumption, the LC-matched EO drive keeps

TABLE I. Simulated ideal operating points (upper bounds; see Table IV and Sec. V for realistic conditions) at $\gamma/2\pi = 0.6$ GHz, $\kappa_i/2\pi = 80.75$ MHz (data/results_B.json) and the measured EO shifter of Ref. [9].

	AO (sim.)	EO ^a (sim.)	Hu [9] (meas.)
$\Omega_m/2\pi$ (GHz)	3.33	3.33	28.2
$\gamma/2\pi$ (GHz)	0.60	0.60	5.31
$\kappa_i/2\pi$ (MHz)	80.75	80.75	170
η_{shift}	98.3%	99.6%	98.7%
η_{abs}	55%	56%	—
IL (dB)	2.5	2.5	0.45
Carrier (dB) ^g	−23.2	−26.0	—
Largest spur (dB) ^g	−21.2	−30.6	—
P_{RF} , as-built feed	6.7 μW ^b	3.1 mW ^c	96 mW
P_{RF} , matched feed	3.4 μW ^d	1.2–13 μW ^e	—
RF bandwidth	~ 1 MHz	≈ 122 MHz ^f	> 3 GHz

^aPush-pull, same optics. ^bShao IDT ($\approx 50\%$ coupling). ^c $V_c = V_0$.

^dIdeal IDT. ^eLC match, ideal inductor to $Q_L = 30$. ^fLC match with $Q_L = 30$; the unmatched electrode is broadband. ^gRelative to the shifted line.

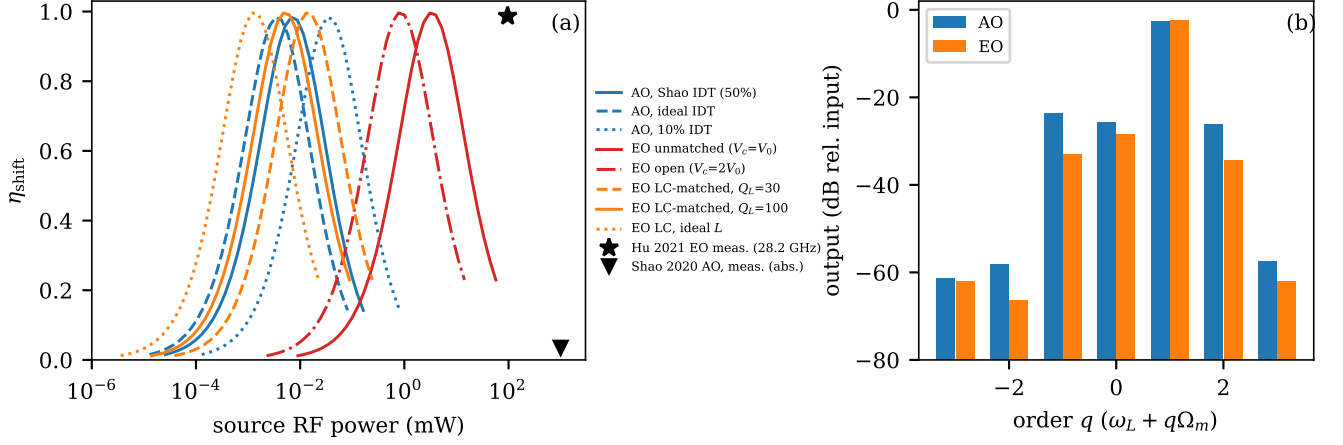


FIG. 2. Simulated operating point ($\gamma/2\pi = 0.6$ GHz, $\kappa_i/2\pi = 80.75$ MHz). (a) η_{shift} versus source RF power for the acoustic drive (blue: Shao IDT with $\approx 50\%$ coupling, ideal IDT, 10% IDT) and for the EO drive with identical optics (red: unmatched, $V_c = V_0$, and open capacitor, $V_c = 2V_0$; orange: LC matched with inductor $Q_L = 30$, $Q_L = 100$, and an ideal inductor). Symbols mark measured devices: the EO shifter of Ref. [9] (star; 98.7% at 96 mW and 28.2 GHz) and the acousto-optic shifter of Ref. [12] (triangle; 3.5% on-chip efficiency at 1 W, which is an absolute rather than a shift-ratio efficiency). (b) Output power in each order q relative to the input at the respective optima of the AO and EO drives.

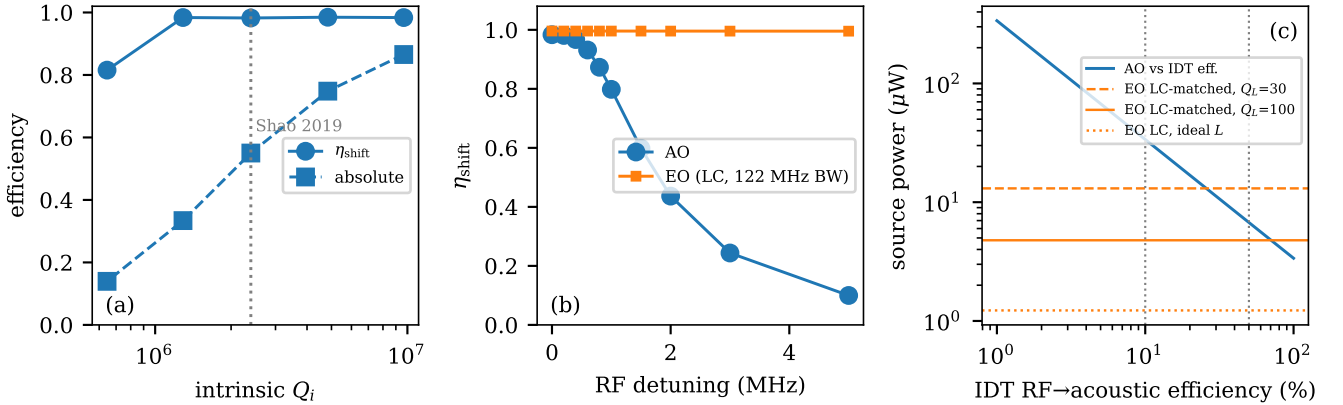


FIG. 3. Tolerances of the acoustic design (simulation). (a) η_{shift} (circles) and η_{abs} (squares) versus intrinsic Q_i , with $\delta\omega$ re-optimized at each point; the dotted line marks the Q_i implied by Ref. [13]. (b) η_{shift} versus RF detuning of the drive from the doublet splitting. For AO the drive amplitude follows the acoustic Lorentzian ($Q_m = 2000$, FWHM 1.67 MHz); the EO curve assumes a flat drive within the 122-MHz bandwidth of the $Q_L = 30$ LC match. (c) Source power of the acoustic drive versus IDT RF-to-acoustic efficiency, compared with the LC-matched EO drive (horizontal lines). Dotted vertical lines mark 10% and 50%.

$\eta_{\text{shift}} = 99.6\%$ over the full ± 5 MHz range.

IDT efficiency.—The acoustic source power scales as $1/\eta_{\text{IDT}}$ [Fig. 3(c)]: $3.4 \mu\text{W}$ at 100%, $6.7 \mu\text{W}$ at the Shao IDT's $\approx 50\%$, and $34 \mu\text{W}$ at 10%, a value reported for other LN acousto-optic microrings [19]. The acoustic drive needs more power than the $Q_L = 30$ LC-matched EO drive ($13 \mu\text{W}$) once $\eta_{\text{IDT}} \lesssim 25\%$, more than the $Q_L = 100$ match ($4.8 \mu\text{W}$) below $\approx 70\%$, and more than the ideal-inductor match ($1.2 \mu\text{W}$) for any physical $\eta_{\text{IDT}} \leq 100\%$.

Thermal drift and ring mismatch.—We extend Eq. (1) with a common laser-ring detuning Δ , a resonance mis-

match δ_r of ring 2 relative to ring 1, and a splitting error $2\mu - \Omega_m$, keeping the drive at the operating point (Fig. 4); at zero offset we recover $\eta_{\text{shift}} = 98.3\%$. We find $\eta_{\text{shift}} \geq 95\%$ for $-72 \leq \Delta/2\pi \leq +55$ MHz ($\geq 90\%$ for -97 to $+85$ MHz), for $-104 \leq \delta_r/2\pi \leq +145$ MHz, and for $-96 \leq (2\mu - \Omega_m)/2\pi \leq +53$ MHz. Within the $\geq 95\%$ region the IL stays between 2.50 and 2.66 dB. To convert to temperature we use $|df/dT| \approx f(dn_e/dT)/n_g \approx 2.8$ GHz/K, with $f = 193.4$ THz and $dn_e/dT = 3.34 \times 10^{-5} \text{ K}^{-1}$ at 1523 nm [20]. With the thermo-optic effect alone, a common drift must be held within $\approx \pm 20$ mK (± 30 mK for $\eta_{\text{shift}} \geq 90\%$)

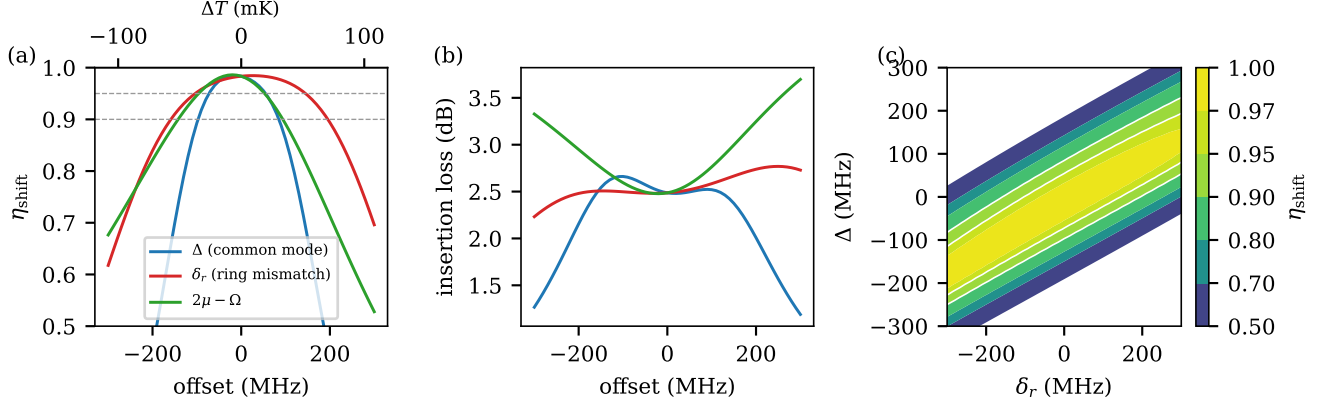


FIG. 4. Thermal-drift, ring-mismatch, and splitting tolerances of the acoustic design at the AO operating point (simulation; `data/tol_scan.csv`). η_{shift} versus the common detuning Δ of the laser from both rings, the ring-2–ring-1 resonance mismatch δ_r , and the splitting error $2\mu - \Omega_m$; the two-dimensional map shows the ridge along $\Delta \approx \delta_r/2$. Temperature axes use $|df/dT| \approx 2.8$ GHz/K (thermo-optic effect only); including thermal expansion (≈ 5.6 GHz/K) halves all temperature windows (see text).

TABLE II. Source RF power at the optimum η_{shift} (simulation; `data/results_B.json`, `data/eff_vs_power.csv`).

Drive and feed	P_{RF}
AO, Shao IDT ($\approx 50\%$)	$6.7 \mu\text{W}$
AO, ideal IDT (100%)	$3.4 \mu\text{W}$
AO, 10% IDT	$34 \mu\text{W}$
EO, unmatched ($V_c = V_0$)	3.1 mW
EO, open capacitor ($V_c = 2V_0$)	0.77 mW
EO, LC match, $Q_L = 30$	$13 \mu\text{W}$
EO, LC match, $Q_L = 100$	$4.8 \mu\text{W}$
EO, LC match, ideal inductor	$1.2 \mu\text{W}$
Hu <i>et al.</i> [9] (meas., 28.2 GHz)	96 mW

and the differential ring temperature within ± 37 mK. Thermal expansion, $\alpha_a \approx 1.54 \times 10^{-5} \text{ K}^{-1}$ [21], shifts the resonances in the same direction; with $|df/dT| \approx f(dn_e/dT + n_{\text{eff}}\alpha_a)/n_g \approx 5.6$ GHz/K ($n_{\text{eff}} = 2.14$ assumed), consistent with the 5.6 GHz/K predicted for e-polarized modes of a bulk LN resonator [22], the windows shrink to ± 9.8 mK (± 15 mK for $\eta_{\text{shift}} \geq 90\%$) for common drift and ± 18.6 mK for the differential temperature. A bonded thin film is partly clamped by the substrate, so the bulk expansion makes these narrower windows conservative; the true values lie between the two estimates. The mismatch is partly absorbed by retuning the laser along the ridge $\Delta \approx \delta_r/2$, and can be removed with a heater on ring 2. The splitting error, $\approx \pm 1.6\%$ of Ω_m , is set by the coupling gap and is thus a fabrication tolerance; the RF frequency can track the actual 2μ only within the IDT and acoustic-resonance bandwidth.

V. REALISTIC CONDITIONS: MONTE CARLO ANALYSIS

The ideal operating point assumes identical rings, perfect temperature control, no fabrication error, and only the measured intrinsic loss of Ref. [13]. To obtain the prediction that we regard as the main result of this study, we repeat the simulation with the parameters of Table IV in a 1000-sample Monte Carlo (Fig. 5). Each sample draws independent ring–ring and ring–bus gap errors (which change μ and γ), an independent residual temperature offset for each ring after locking, and a chip-to-chip spread of Q_i , with $Q_m = 2000$ fixed; the RF frequency is retuned to follow the actual splitting within ± 1 MHz. The acoustic Q_m affects only the drive power and the RF bandwidth, not η_{shift} .

Results.—The median shift ratio is $\eta_{\text{shift}} = 94.3\%$ (10th and 90th percentiles 82.4% and 97.9%); 44.7% of the samples reach $\eta_{\text{shift}} \geq 95\%$ and 73% reach $\geq 90\%$. The median on-chip IL is 5.0 dB (3.9–6.1 dB), dominated by the assumed extra loss, and the median on-chip absolute efficiency is 28.4% (21.5–38.8%). With 1.5 dB per facet the fiber-to-fiber efficiency is 14.2% (10.8–19.4%). At 10% IDT efficiency the source power is $33.7 \mu\text{W}$ at $Q_m = 2000$, $18.7 \mu\text{W}$ at the measured maximum $Q_m = 3600$ (upper bound), and $43 \mu\text{W}$ at $Q_m = 1570$, an assumed conservative value without a literature source (power $\propto 1/Q_m$). These numbers, not the ideal 98.3% at $6.7 \mu\text{W}$, are our central prediction; they remain simulation results.

Dominant risks.—The largest fabrication risk is the coupling gap. A 5-nm gap error produces a splitting error $2\mu - \Omega_m$ of ≈ 114 MHz rms, and in 99% of the samples $|2\mu - \Omega_m|$ exceeds the ± 1 MHz retuning range, so RF retuning compensates almost nothing (without retuning the median is 0.945 and the 10th percentile 0.82 in

TABLE III. Benchmarks. “Meas.” denotes measured devices, “sim.” this work; LNOS, LN-on-sapphire; BW, RF bandwidth. All except Ref. [16] are on TFLN. The efficiencies are not all defined identically (see notes).

Work	Mechanism	Shift	Efficiency	Drive	Note
Shao <i>et al.</i> [12] (meas.)	AO traveling wave	~3 GHz	3.5% on-chip ^a	1 W (30 dBm)	70 MHz BW
Zhu <i>et al.</i> [16] (meas.)	AO backward Brillouin (LNOS)	8.54 GHz	reflectivity $\leq 24\%$	mW (acoustic)	no forward η_{shift}
Hu <i>et al.</i> [9] (meas.)	EO photonic molecule	28.2 GHz	$\eta_{\text{shift}} = 98.7\%$	96 mW	IL 0.45 dB, >3 GHz BW
Qiu <i>et al.</i> [11] (meas.)	EO serrodyne	—	>98.8%	$V_{\pi} \approx 2.3$ V	broadband ^b
This work (sim.) ^c	AO photonic molecule	3.33 GHz	$\eta_{\text{shift}} = 94.3\%$, $\eta_{\text{abs}} = 28.4\%$	33.7 μ W	IL 5.0 dB, ~1 MHz BW
This work, ideal	upper bound	3.33 GHz	$\eta_{\text{shift}} = 98.3\%$, $\eta_{\text{abs}} = 55\%$	6.7 μ W	IL 2.5 dB
EO ref. (sim.)	EO molecule, LC matched	3.33 GHz	$\eta_{\text{shift}} = 99.6\%$	1.2–13 μ W	122 MHz BW ($Q_L = 30$)

^aDeflected power divided by transmitted power with the microwave off [12]. ^bValues taken from the published abstract of Ref. [11].

^cRealistic conditions, Monte Carlo medians (Sec. V).

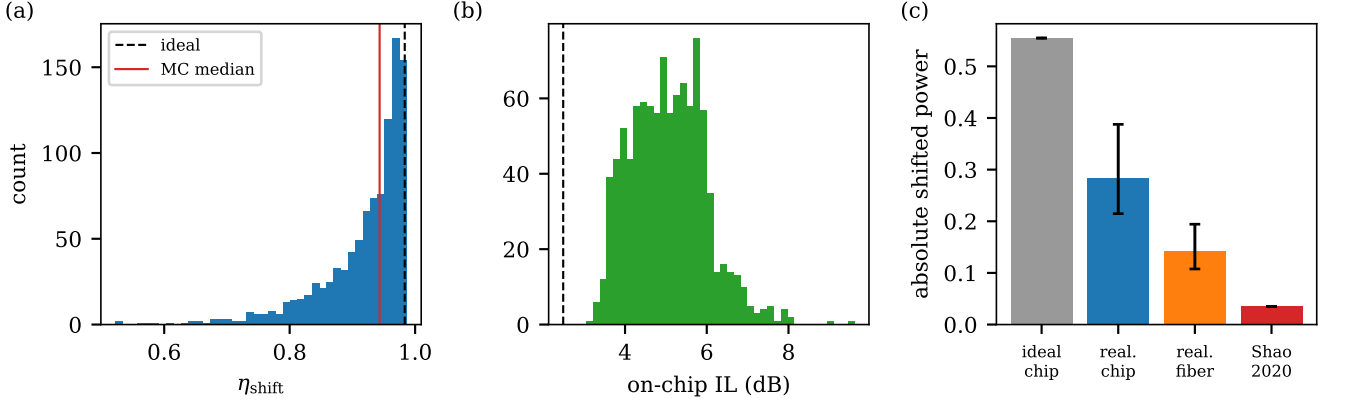


FIG. 5. Realistic-condition Monte Carlo of the acoustic design (simulation, $N = 1000$; `data/mc_samples.csv`, `data/realistic_summary.json`) with the parameters of Table IV: distributions of η_{shift} , on-chip IL, and on-chip and fiber-to-fiber absolute efficiency, together with one-at-a-time sensitivity scans of the assumed parameters (each scan point drawn independently with $N = 300$, statistical error $\approx \pm 0.005$ in the median; trends only). The ideal operating point ($\eta_{\text{shift}} = 98.3\%$) is an upper bound.

the $N = 300$ sensitivity scan, essentially unchanged and within its $\approx \pm 0.005$ statistical error of the $N = 1000$ median 0.943). Matching the splitting to Ω_m therefore relies on gap precision within ≈ 5 nm; with $\sigma = 10$ nm the median falls to 0.89 (10th percentile 0.64), and with 20 nm to 0.75. The results are also sensitive to the residual lock jitter: the median is 0.95 at 1 mK, 0.91 at 10 mK, and 0.80 at 20 mK (10th percentile 0.28). The extra loss mainly affects the IL: doubling κ_i leaves the median $\eta_{\text{shift}} \approx 0.95$ but raises the IL from 2.5 to ≈ 4.9 dB, whereas a fivefold κ_i drops the median to 0.54.

VI. SINGLE-PHOTON PERFORMANCE

HOM visibility.—A frequency shifter for quantum networks must preserve the spectral-temporal mode of the photon. We model an input photon with a Gaussian spectral amplitude $\psi(\Delta) \propto \exp(-\Delta^2/4\sigma^2)$, where σ is the rms width of $|\psi|^2$, and compute the complex transfer function $H(\Delta)$ of the shifted ($q = +1$) channel from the full time-domain solution of Eq. (1) at the

AO operating point, evaluated on 41 points over $\pm 4\sigma$ (`data/H_shift.csv`). The shifted photon, filtered to its target frequency bin, interferes with a reference photon of the same spectrum at the target frequency. Because a fixed delay of the reference can always be added, we maximize over the delay τ :

$$V = \max_{\tau} \frac{\left| \int \psi^*(\Delta) H(\Delta) \psi(\Delta) e^{-2\pi i \Delta \tau} d\Delta \right|^2}{\int |\psi|^2 d\Delta \int |H\psi|^2 d\Delta}. \quad (3)$$

The normalization by $\int |H\psi|^2$ assumes that the loss is balanced, e.g. by attenuating the reference arm, so V measures mode distinguishability only. The optimum delay is 1.0–1.1 ns, i.e. the group delay of the device.

The results are shown in Fig. 6. With filtering, $1 - V = 8 \times 10^{-7}$ at $\sigma = 10$ MHz, 3×10^{-5} at 25 MHz, 6×10^{-4} at 50 MHz, 1.1×10^{-2} at 100 MHz, and 9×10^{-2} at 200 MHz. The residual distinguishability comes from the curvature of $H(\Delta)$ beyond linear phase across the photon spectrum, which grows rapidly as σ becomes a sizable fraction of the optical linewidths ($\gamma/2\pi = 0.6$ GHz). Without filtering, the residual carrier and spurious orders limit the visi-

TABLE IV. Parameters of the realistic-condition analysis (data/params_sources.md). “Assumed” marks values without a direct measurement on this platform; each was varied in a separate sensitivity scan (range in parentheses).

Parameter	Value (scan)	Source
IDT efficiency	10% (50% ref.)	50%: calibration of [13]; 10%: conservative, cf. [19]
Acoustic Q_m	2000, fixed (3600 upper; 1570 cons.)	2000: typical value of [13]; 3600: measured max. [13]; 1570: assumed, no source
Extra intrinsic loss (IDT metal, suspension)	$\kappa_i \times 2$ (1–5 $\times$)	assumed
Q_i chip-to-chip spread	$\kappa_i \times U(0.7, 1.3)$	assumed
dn_e/dT	$3.34 \times 10^{-5} \text{ K}^{-1}$	[20]
α_a	$1.54 \times 10^{-5} \text{ K}^{-1}$	bulk [21]
Residual lock jitter	5 mK rms (1–20 mK)	assumed
Coupling-gap error σ	5 nm (1–20 nm)	assumed
Coupling decay length	150 nm	assumed
RF retuning range	± 1 MHz (0, 1, 10, 35 MHz)	derived: about half the acoustic FWHM (1.67 MHz at $Q_m = 2000$)
Fiber-chip coupling	1.5 dB/facet (0.5–3 dB)	meas. edge couplers [23, 24]

bility to $V \approx \eta_{\text{shift}}$: $V = 0.9835$ at $\sigma = 10$ MHz and 0.9829 at 50 MHz, i.e. $1 - V \approx 1.7 \times 10^{-2}$, more than four orders of magnitude worse than the filtered value at 10 MHz ($V = 0.9999992$). A spectral filter on the target bin is therefore required for high-visibility operation. The device is therefore suited to narrowband photons with $\sigma \lesssim 50$ MHz, such as cavity-enhanced parametric sources or atom- and quantum-dot-like emitters, together with a spectral filter on the target bin. Broadband photons require a wider optical linewidth, i.e. a larger γ and correspondingly larger drive.

Thermal phonons.—The shift is a pump-free, number-conserving, beam-splitter-type process: the acoustic field scatters a photon from the S to the AS supermode, and no photon is created without an input photon. Thermal occupation n_{th} of the acoustic mode therefore adds only an incoherent fluctuation of the coupling, of relative size $n_{\text{th}}/N_{\text{coh}}$, where $N_{\text{coh}} = (4\gamma_e/\gamma^2) P_{\text{RF}}/\hbar\Omega_m$ is the coherently driven phonon number. With $\gamma = \Omega_m/Q_m$, $Q_m = 2000$, and $\gamma_e/\gamma = 0.15$ [13], we find $N_{\text{coh}} \approx 1.75 \times 10^{11}$ at the operating point. At 3.33 GHz, $n_{\text{th}} = 1877$ at 300 K, 24.5 at 4 K, and 1.1×10^{-7} at 10 mK, so $n_{\text{th}}/N_{\text{coh}} = 1.1 \times 10^{-8}$, 1.4×10^{-10} , and 7×10^{-19} , respectively. Thermal-phonon noise is therefore negligible even at room temperature, unlike microwave-to-optical transduction, in which thermal phonons are converted into noise photons. We did not model spontaneous scattering

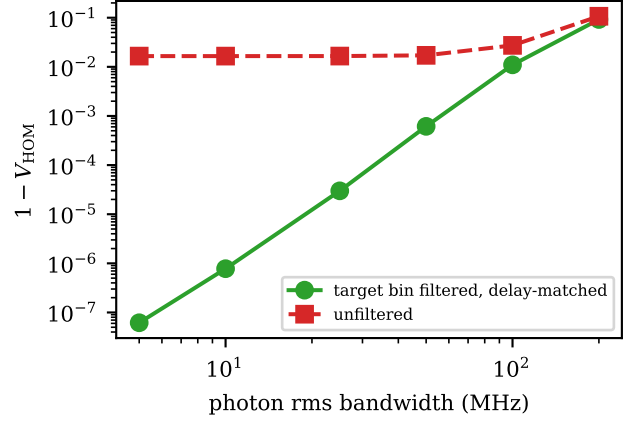


FIG. 6. HOM infidelity $1 - V$ versus photon rms bandwidth σ (simulation; data/HOM.csv). Circles: shifted photon filtered to the target bin, reference photon delay-matched [Eq. (3)]. Squares: unfiltered, where the residual unshifted and spurious orders limit V to $\approx \eta_{\text{shift}}$.

of photons into other optical modes by thermal phonons; with the single-phonon coupling $g_0/2\pi \approx 1.1$ kHz of this platform [13] we expect it to be much smaller than the effects above, but this remains to be quantified.

VII. LIMITATIONS

The results above are design predictions subject to the following limitations.

(i) *No power advantage over matched EO.* With impedance matching on both sides the acoustic and EO source powers differ only by a factor of 0.36–3.9, and the LC-matched EO drive is more power-efficient with a low-loss inductor (Sec. III). The advantage claimed here is restricted to the class of acousto-optic frequency shifters.

(ii) *Narrow RF bandwidth and fixed shift frequency.* The usable RF bandwidth is ~ 1 MHz, set by the acoustic $Q_m = 2000$, versus ≈ 122 MHz for a $Q_L = 30$ LC-matched EO electrode and >3 GHz for the unmatched device of Ref. [9]. The shift is fixed at the 3.33-GHz acoustic resonance, whereas EO photonic molecules have been operated at 12.5–28.2 GHz [9]. The RF source must be frequency-stabilized, and the doublet splitting must be thermally tuned to match Ω_m to well within the optical linewidth. Maintaining $\eta_{\text{shift}} \geq 95\%$ requires active stabilization of the common resonance at the ± 9.8 mK level when thermal expansion is included (± 20 mK for the thermo-optic effect alone; Sec. IV), e.g. with heater electrodes or by locking the laser to the resonance; passive temperature control is insufficient. The ring-ring mismatch, in contrast, can be corrected with a heater and is not a fundamental limit.

(iii) *Insertion loss.* The ideal IL is 2.5 dB ($\eta_{\text{abs}} = 55\%$), limited by the measured $Q_i \approx 2.4 \times 10^6$ of the suspended

platform [13]; under realistic conditions the median is 5.0 dB, compared with 0.45 dB for the measured EO device [9]. The fiber-to-fiber estimate assumes 1.5 dB per facet, which requires an optimized edge coupler [23]; the suspended chip of Ref. [13] had $\lesssim 5$ dB per facet.

(iv) *IDT efficiency.* The acoustic drive loses to a $Q_L = 30$ LC-matched EO drive if the IDT efficiency is below $\approx 25\%$; IDT efficiencies of $\approx 10\%$ have been reported in related LN microring devices [19].

(v) *Drive extrapolation.* We obtain $\delta\omega$ by extrapolating the measured MZI $V_\pi L$ of Ref. [13] linearly to a ring that the acoustic field covers only partially (15%). We assume the same IDT efficiency, mode overlap, and uniform acoustic phase along the covered section as in the MZI. This extrapolation must be verified experimentally; $n_g = 2.3$ is assumed.

(vi) *Fabrication and locking (Sec. V).* The splitting 2μ must match Ω_m ; a 5-nm gap error already gives ≈ 114 MHz rms splitting error, far beyond the ~ 1 -MHz acoustic linewidth. RF retuning is limited to ± 1 MHz by the acoustic resonance and leaves 99% of the samples uncompensated, so the coupling gap is the largest fabrication risk of the design: gap precision within ≈ 5 nm is required, and at 10 nm the median η_{shift} falls to 0.89 (10th percentile 0.64). A heater on the inter-ring coupler could make the splitting tunable; we suggest this but have not modeled it. The result is also sensitive to the residual lock jitter (median 0.91 at 10 mK, 0.80 at 20 mK).

(vii) *Assumed values.* The extra optical loss, residual lock jitter, gap error, coupling decay length, Q_i spread, and the conservative acoustic $Q_m = 1570$ (no literature source) are assumptions without direct measurements on this platform (Table IV); we report sensitivity scans for each. The extra-loss term is possibly double counted, because the measured Q_i of Ref. [13] already comes from a suspended device with an IDT, so the realistic IL is likely pessimistic. The bulk thermal-expansion coefficient likely overestimates the expansion of a bonded film.

(viii) *Not modeled.* We did not model ohmic and acoustic heating of the IDT and resonator, the resulting drive-dependent drift of the ring resonances beyond the lock jitter above (Ref. [13] observed a heating-induced red shift with increasing microwave power), or the mechanical stability of the suspended structure.

(ix) *Model provenance.* Our implementation of Eq. (1) is reconstructed from the published equations of Ref. [9],

not from the authors' code, and the design combines parameters of two different devices (optics and electrodes from Ref. [9], acoustics and loss from Ref. [13]). Any extension that routes the acoustic drive through on-chip phononic circuits (for example, to feed several shifters) would at present rely on acoustic-circuit parameters borrowed from GaN-on-sapphire [25] and on our reconstruction of that model from its equations; none of the numbers reported here depend on it.

(x) *Simulation only.* No device has been fabricated; every number in this paper, including the realistic Monte Carlo, is a model prediction.

VIII. CONCLUSION

In simulation, a suspended IDT acoustic resonator driving one ring of a TFLN photonic molecule shifts light by 3.33 GHz with a median shift ratio of 94.3% (10th–90th percentile 82.4–97.9%), a median on-chip absolute efficiency of 28.4% (5.0 dB IL), and 14.2% fiber to fiber, at 33.7 μW of source power ($Q_m = 2000$) with a 10%-efficient IDT, under realistic and partly assumed conditions. The ideal operating point, 98.3% at $\approx 7 \mu\text{W}$ with 55% absolute efficiency, is an upper bound. For reference, the best measured acousto-optic shifter on TFLN, a broadband (70 MHz) traveling-wave device, reaches 3.5% on-chip at 1 W [12], versus $\approx 28\%$ on-chip at $\approx 34 \mu\text{W}$ here; our design is resonant and simulated, with a ~ 1 -MHz RF bandwidth and a fixed shift frequency. Unlike the backward-Brillouin photonic molecule of Ref. [16], it operates as a forward, single-bus shifter. It is not more power-efficient than an impedance-matched EO drive of the same optics (factor 0.36–3.9). Reaching the predicted performance requires resonance locking at the ± 9.8 mK level and coupling-gap control within ≈ 5 nm, the largest fabrication risk. With a delay-matched reference and filtering of the target bin, the HOM infidelity at the ideal point is 8×10^{-7} for 10-MHz and 6×10^{-4} for 50-MHz photons, and thermal phonon noise is negligible at room temperature. The decisive experiments are a measurement of the acousto-optic coupling of a partially covered high- Q ring, the IDT efficiency and the loss and heating added by the IDT, and the achievable gap control and lock stability.

Data availability.—Simulation scripts (`core.py`, `make_all.py`, `hom_post.py`) and data files are available from the author.

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